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Three-photon neuronal imaging in deep mouse brain

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Three-photon microscopy has been increasingly adopted for probing neural activities beyond the typical two-photon imaging depth. In this review, we outline the unique properties that differentiate three-photon microscopy from two-photon microscopy for *in vivo* imaging in biological samples, especially in the mouse brain. We present a systematic summary of the optimization of three-photon imaging parameters for neural imaging, based on their effects on calcium imaging quality and perturbation to brain tissues. Furthermore, we review the existing techniques for volumetric imaging and discuss their prospects in mesoscale three-photon imaging in deep tissue. © 2020 Optical Society of America under the terms of the OSA Open Access Publishing Agreement

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1. INTRODUCTION

Two-photon microscopy (2PM) has revolutionized *in vivo* functional imaging in the last 30 years—enabling high-spatial-resolution, neurophysiological recording at an unprecedented scale and depth inside the scattering brain [1]. Although the concept of three-photon microscopy (3PM) was conceived and demonstrated as early as the mid-1990s [2–4], the advantages of 3PM were not fully understood initially. The required higher excitation pulse energy and longer excitation wavelength for 3PM when compared to 2PM, coupled with the concern for tissue heating, severely dampened the enthusiasm for the subsequent development of 3PM [5]. Other than expanding the wavelength accessibility of the femtosecond laser [6], applications of three-photon excited fluorescence for deep tissue imaging were not actively pursued until the 2010s. Two discoveries in 2013 underpin the new excitement for *in vivo* 3PM: (1) the optimal excitation wavelengths for deep brain imaging are around 1300 nm and 1700 nm [7], which are too long for two-photon excitation (2PE) of most existing fluorescent probes, and (2) the higher-order nonlinear excitation provides orders of magnitude higher excitation confinement when compared to 2PE, which is necessary for high-contrast imaging deep in scattering biological tissues [7]. In the first experiment for *in vivo* deep brain imaging, 3PM with 1700 nm excitation was used to image vascular and neuronal structures at ∼1.3 mm depth in an intact mouse brain through the scattering white matter [7]. The rapid development of 3PM for deep brain imaging since then was propelled by the progress in genetically engineered calcium indicators [8] and the commercial availability (starting ∼2016) of energetic and robust femtosecond laser sources for three-photon excitation (3PE) at the long-wavelength windows. Several research groups have since adopted the three-photon (3P) imaging technique and have demonstrated its capability for probing neural activities beyond the typical depth limit of two-photon (2P) imaging. 3PM of neuronal activity was first demonstrated in the hippocampus of intact adult mouse brains [9]. 1300 nm 3PM was also found to be capable of imaging neuronal activity with single-cell resolution through the intact skull of awake, adult mice [10]. 3PM was applied to neurophysiological studies on evoked neuronal responses to the drifting visual gratings throughout the entire column, including the subplate of the mouse visual cortex [11–13]. Multiplexed 2P and 3P volumetric imaging technology was further developed that combines the advantages of 2P and 3P for simultaneous monitoring of both shallow and deep cortical layers (auditory cortex, parietal cortex, etc.) as well as subcortical regions in awake mice [14]. Recently, a head-mounted miniature 3PM was demonstrated for recording neuronal activities at >1.1 mm depth in the posterior parietal and visual cortex of freely moving rats [15]. Long-wavelength 3PM is poised to play a major role in deep imaging within scattering biological tissues, and it has potential impacts in a large number of biomedical fields such as neuroscience, immunology, cancer biology, etc. In this paper, we review the state-of-the-art 3P imaging technology and provide some perspective on its future for large-scale activity recording in the deep brain.

2. THREE-PHOTON VERSUS TWO-PHOTON NEURONAL IMAGING

In this section, we provide a comprehensive comparison between 3PM and 2PM for deep tissue imaging. The result of this comparison is summarized in Fig. 1 and provides clear guidelines on the

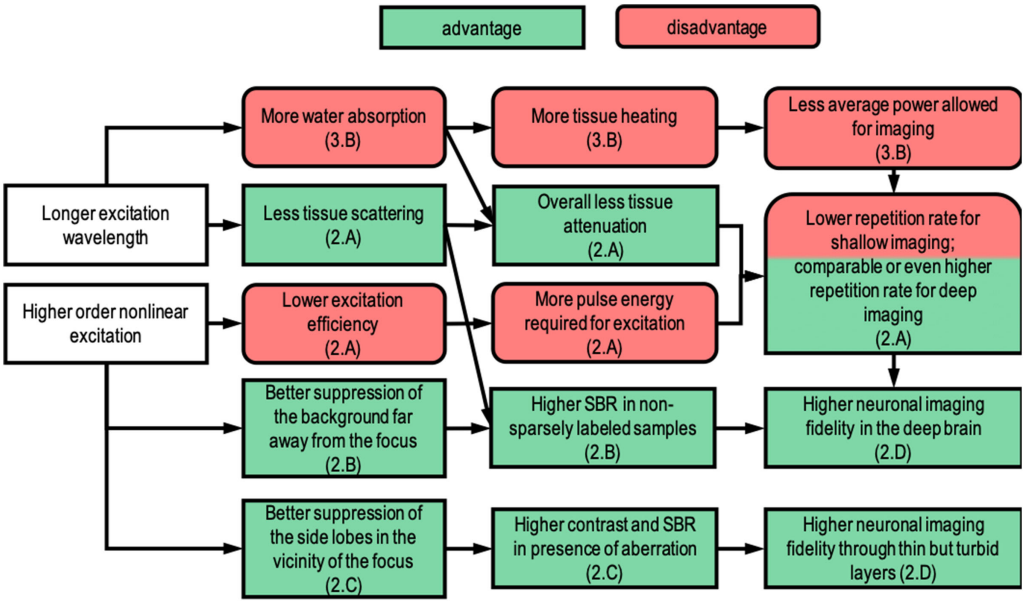


Fig. 1. Cause-and-effect diagram on the imaging properties resulted from the long-wavelength 3PE, in comparison to 2PE of the same fluorophores. The topics are discussed in detail in the subsections indicated in each box.

choice of the imaging modality (i.e., 3PM or 2PM) under different imaging conditions.

A. Long-Wavelength 3PM Has Higher Signal Generation Efficiency for Deep Imaging

The signal generation efficiency of deep brain 3P imaging benefits substantially from the reduced tissue attenuation of the long excitation wavelength. Although 3PE requires higher excitation intensity at the focus (∼5−10 times) to generate the same amount of fluorescence as 2PE [17–19], for deep brain imaging, tissue attenuation of ballistic photons is the dominant factor that determines the total required imaging power. The longer wavelengths used for 3PE around 1300 nm and 1700 nm experience far less tissue scattering compared to the shorter wavelengths commonly used for 2PE, and they are at the local minima of the water absorption spectrum to reduce tissue absorption [Fig. 2(A)] [7,16]. Accounting for both the scattering and absorption effects, 3PE light experiences less overall tissue attenuation than 2PE, characterized by a much longer effective attenuation length (EAL), which is defined as the propagation distance at which the ballistic photons are attenuated by $1/e$ [7]. The EAL determines the rate at which excitation power grows *exponentially* with imaging depth in order to keep the fluorescence signal constant at the focus. For example, the EAL at 1300 nm in the mouse neocortex is approximately twice that of 920 nm for 2PE (∼300 μm versus ∼150 μm in the somatosensory cortex) [17], which means 3PE power attenuation is e times less than 2PE for every 300 μm increment of imaging depth (i.e., for every 3P EAL). As a result, 3PE experiences 10 times less attenuation at ∼690 μm imaging depth (∼2.3 3P EALs) than 2PE, more than enough to compensate for the higher excitation pulse energy required for 3PE of GCaMP6 [17]. This depth is defined as the *signal crossover depth* where 3PE generates an equal amount of signal per excitation pulse as 2PE at a shorter wavelength, given the same fluorophore, pulse energy at the brain surface, and pulse duration [Figs. 2(B) and 2(C), top panels] [20]. With the same maximum allowable average powers, the signal crossover depth indicates the imaging depth where 2PM and 3PM would have the same maximum repetition rate. As the imaging depth increases beyond the crossover depth, the maximum repetition rate for 2PM will be below that of 3PM in order to keep the signal the same. In practice, short-wavelength 2PM permits higher average power than long-wavelength 3PM, and this topic will be revisited in the discussion on imaging power in Section 3.B.

B. Long-Wavelength 3PM Has a Higher Signal-to-Background Ratio for Deep Imaging

The out-of-focus background in multiphoton imaging has two main components. The bulk background refers to the fluorescence generation in the light cone away from the focus, and the defocus background refers to the fluorescence generated by the side lobes of a distorted point-spread function (PSF) [Fig. 3(A)]. Both types of background lead to loss of contrast in the images and contribute to the total fluorescence generation outside the intended diffraction-limited focal volume. However, the bulk background depends strongly on the normalized imaging depth and labeling density, and it exists even in the absence of any PSF distortion. On the other hand, the defocus background can exist at a shallow imaging depth and with sparse labeling if the focus is distorted by strong tissue aberration. In practice, the background observed in the captured images is often a mixture of the two, and their relative importance depends on imaging depth, labeling density, and the optical properties of the tissue [Figs. 3(B) and 3(C)]. 3PM helps to suppress both types of out-of-focus background by using longer wavelength and higher-order nonlinear excitation (Fig. 1). The bulk background is discussed in this section, and the defocus background is discussed in Section 2.C.

The imaging depth of 2PM is fundamentally limited by the background fluorescence in non-sparsely labeled samples. In the absence of severe aberration, i.e., imaging with a PSF with negligible side lobes, the primary contributor to the background is the out-of-focus multiphoton excitation in the sample away from the focus [Fig. 3(A) left panel and 3(B)] [22], which appears

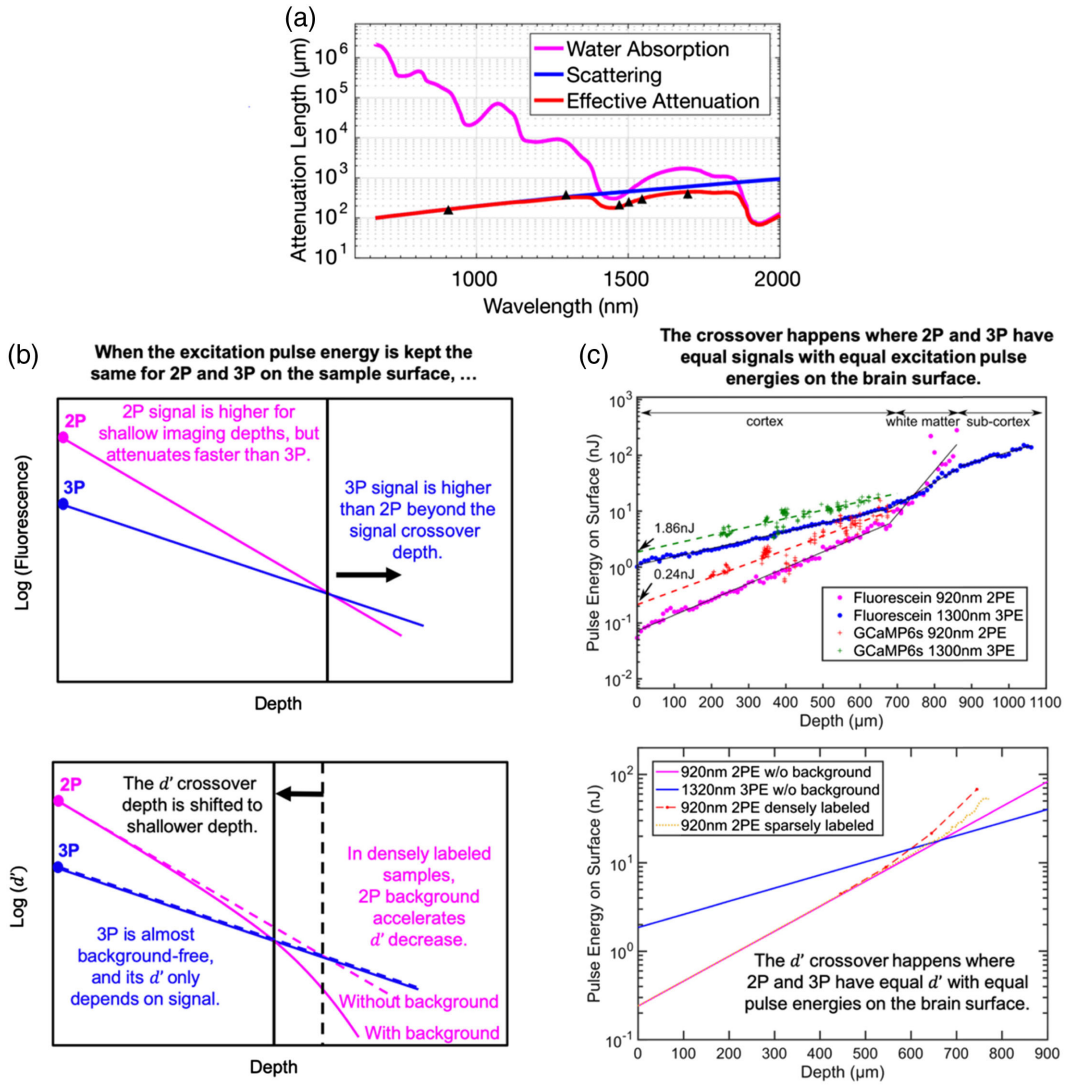


Fig. 2. Wavelength dependence of mouse brain effective attenuation length and its impact on two- and three-photon excitation signal crossover for deep imaging. (A) The theoretical model of the effective attenuation length based on water absorption and Mie scattering. The black triangles indicate the measured effective attenuation lengths in mouse brains *in vivo*. Reproduced with permission from Ref. [16], 2018, © The Optical Society. (B) Schematic illustration of the signal and d' crossover between 2PM and 3PM in the same sample, with the same pulse energy on the sample surface (i.e., the same average power and repetition rate). (C) Measurement data for the signal crossover depth in the mouse brain (top) and the d' crossover depth in the presence of background (bottom). Reproduced with permission from Ref. [17], 2020, eLife Sciences Publications.

as a featureless background in images and grows rapidly with depth [Fig. 3(B)]. The strength of the bulk background primarily depends on the normalized imaging depth (i.e., the physical depth divided by the EAL) and the ratio of the staining density in the focal volume to that in the out-of-focus volume [22,23]. When the background becomes comparable to the signal generated from the focus, images suffer from low contrast and high background shot noise, which causes unrecoverable loss of spatial and temporal information. Although the power-squared dependence of 2PE can effectively reduce the out-of-focus excitation in scattering samples, its imaging depth is still limited to ∼5 EALs for a labeling density of ∼2% (e.g., brain vasculature), where the signal-to-background ratio (SBR) approaches one [22,24,25]. The SBR limit has been experimentally observed to occur at 450 ∼ 850 μm in the mouse cortex with ∼920 nm excitation, depending on the sample variation and labeling density [12,17,24]. Despite several reports on 2P imaging in the deep cortex or even the hippocampus of the mouse brain, these studies imposed additional requirements to reduce the labeling density or the number of EALs, such as layer-specific staining with redshifted dyes [26], the removal of the neocortex [27–30], or imaging young mice with more transparent brains [31]. Nevertheless, none of these measures fundamentally extends the depth limit imposed by the SBR on 2PM.

1300 nm 3PM is free of background generation for most practical imaging depths, benefited from both the longer excitation wavelength and the higher-order nonlinearity. Since the SBR of multiphoton microscopy primarily depends on the normalized depth, the long excitation wavelength substantially improves the SBR by reducing the number of EALs at the same physical depth. Experiments have shown that 1300 nm 3PM is essentially background-free throughout the entire mouse cortex (up to ∼800 μm) [12,17], which is less than three EALs for 1300 nm but ∼6 EALs for 920 nm. Furthermore, the higher order of nonlinear excitation (I^3 versus I^2) also plays a critical role in background

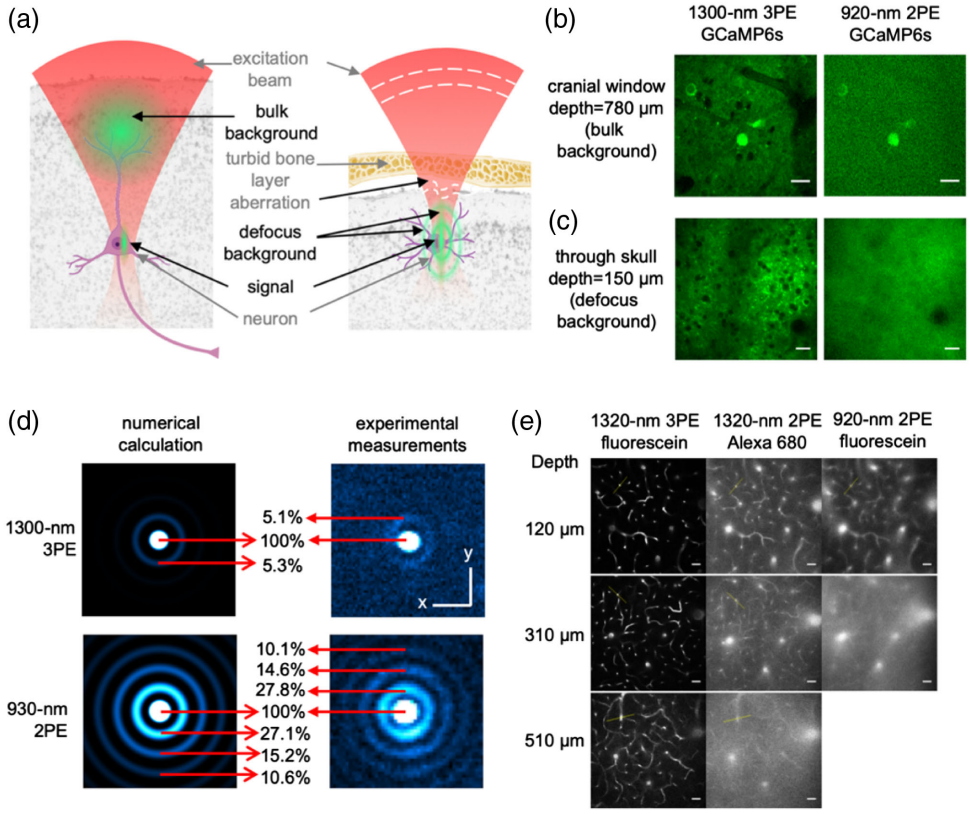


Fig. 3. Background reduction by 3PE. (A) Illustrations of the two types of background encountered by multiphoton imaging in deep tissue or through turbid layers. (B) Demonstration of the bulk background by comparing 920 nm 2PM and 1300 nm 3PM for neuron imaging at the same location in the deep cortical layers (at 780 μm below the brain surface) in a transgenic mouse brain (CamKII- tTA/tetO-GCaMP6s) through a cranial window. Scale bar, 20 μm. Reproduced with permission from Ref. [9], 2017, Springer Nature. (C) Demonstration of the defocus background by comparing 920 nm 2PM and 1320 nm 3PM for neuron imaging at the same location through the intact skull (~150 μm deep in the brain, with an additional of ~105 μm skull thickness above the brain surface) in a transgenic mouse (CamKII- tTA/tetO-GCaMP6s). Scale bar, 20 μm. Reproduced with permission from Ref. [10], 2018, Springer Nature. (D) 3PE significantly reduces the side lobes of a Bessel beam compared to 2PE. Reproduced with permission from Ref. [21], 2018, © The Optical Society. (E) Through skull imaging showing 3PE preserves both the lateral and axial resolution, by comparing 2PM and 3PM images with the same excitation wavelength of 1320 nm. The imaging depth in the brain excludes the thickness of the intact skull. Reproduced with permission from Ref. [10], 2018, Springer Nature.

suppression. In contrast to 2PM, which reaches the SBR limit at ~5 EALs for imaging mouse brain vasculature [17,22,24], 3PM achieves an SBR of >40 at ~2100 μm, which corresponds to more than five EALs, as demonstrated by imaging quantum-dot-labeled vasculature with 1700 nm 3PM [32]. It is noteworthy that the 3P imaging depths is more often limited by signal strength instead of SBR. In order to achieve >2 mm imaging depth in the mouse brain in Ref. [32], quantum dots with 3PE cross sections of 10^4 to 10^5 times that of Texas Red was used.

C. 3PE Has Stronger Excitation Confinement in Turbid Media

The higher-order nonlinearity of 3PE maintains the high SBR and resolution despite the strong aberration caused by the sample [Fig. 3(A) right panel and Fig. 3(C)]. This phenomenon is best demonstrated with 3P Bessel beam excitation because of the characteristic side lobes of Bessel beams. Both simulation and experiment show that 3PE accentuates the central peak and suppresses the side lobes of the PSF, resulting in only ~8.7% of the total fluorescence generated in the side lobes, a significant reduction from the ~73.1% by 2PE [Fig. 3(D)] [33]. The confinement of 3PE has been observed to preserve the resolution and contrast of images in the presence of system- or sample-induced wavefront distortion, which causes severe PSF degradation, especially in the axial direction [34]. For example, 1300 nm 3PM is capable of imaging through the unthinned, intact mouse skull with a high SBR close to 100, while 2PE, even with the same excitation wavelength of 1300 nm, suffers from the overwhelming background with a low SBR of 3 at the same depth [Fig. 3(E)] [10]. Similar effects have been observed with 2PM and 3PM when imaging inside the mouse cranial bones and Drosophila brains [35–37]. It is worth mentioning that the preservation of resolution by 3PE comes at the cost of the substantially reduced excitation power efficiency, which makes imaging at a large depth challenging if the side lobes of the PSF contains a large portion of excitation power [10,38]. One possible way to recover some of the lost excitation efficiency is to use adaptive optics (AO) to correct the wavefront, albeit with added system complexity. When using sensorless, iterative approaches for AO, 3PE facilitates the convergence of wavefront correction algorithms based on the signal strength, since the higher-orderer nonlinear dependence of 3PE signal on the excitation intensity results in a steeper gradient descent for maximizing the signal [37,39–41].

D. 3P Ca^{2+} Imaging is Advantageous in Deep Brain

The advantages of 3PM in background suppression and signal generation efficiency both contribute to its superior calcium imaging (Ca^{2+} imaging) quality in the deep brain. The influences of SBR and signal strength on neuronal activity imaging can be summarized with the discriminability index d' [17,42], which characterizes the detection fidelity of calcium transients:

$$d' \approx \frac{1}{\sqrt{1+1/\mathrm{SBR}}}\frac{\Delta F}{F}\sqrt{\frac{F_0\tau_{\mathrm{Ca}}}{2}}, \tag{1}$$

where τ_{Ca} is the $1/e$ decay time of the calcium indicator, $\Delta F/F$ is the peak fluorescence change of a single-action-potential-induced calcium transient, and F_0 is the baseline brightness of the neuron measured in photons per second. A higher d' indicates more accuracy in calcium transient detection. While 3PM is essentially background-free at practical imaging depths (i.e., $1/\mathrm{SBR} << 1$), 2PM suffers from a reduced effective $\Delta F/F$ by a factor of $(1+1/\mathrm{SBR})^{-1}$ [17]. In addition to the background generation, 2PE also suffers from a lower signal generation efficiency when imaging beyond the signal crossover depth, which results in a lower F_0 for the same excitation power.

Analogous to the signal crossover depth, a d' crossover depth can be defined to quantify the depth beyond which 3PM outperforms 2PM for Ca^{2+} imaging. With the same average power, pulse duration, and repetition rate at the brain surface, 1320 nm 3PE of GCaMP6-labeled neurons generates more signal and therefore d' than 920 nm 2PE at a depth greater than ~690 μm in the mouse cortex [Figs. 2(B) and 2(C) top panels]. When the background is also taken into account, the d' crossover depth shifts to ~600 μm in densely labeled transgenic mouse brains (e.g., CamKII-tTA/tetO-GCaMP6s) [Figs. 2(B) and 2(C) bottom panels] [17].

The higher d' of 3PM ultimately enables more accurate and unbiased neural recording. For example, the low SBR caused by 2PE in the deep brain is particularly detrimental for measuring the neuronal activity of the neurons with low expression levels (i.e., the dimmer neurons). Visual stimulation experiments have shown that the detection accuracy of the neuron orientation tuning by 2PM dropped drastically at the cortical depth of 600 μm, with only ~25% of the neurons showed orientation preference matching that of the 3PM recording [12].

3. OPTIMIZATION OF THREE-PHOTON IMAGING PARAMETERS

The optimization of 3P imaging requires proper quantification of imaging parameters and careful study of their impact on imaging quality and sample physiology. Equations (2) and (3) give the 2PE and 3PE signal per excitation pulse for a Gaussian–Lorentzian focus in an infinite sample (i.e., the sample thickness is much greater than the Rayleigh length of the focused beam) [17,18]:

$$\begin{aligned}\text{2PE signal photons per pulse} &= \frac{S_{2\mathrm{P}}}{f}\\ &= \frac{1}{2}\frac{g_p^{(2)}}{\tau}\phi C(\eta\sigma_2)n_0\frac{\pi}{\lambda_{2\mathrm{P}}}\left(\frac{P_{2\mathrm{P}}}{f}\right)^2,\end{aligned} \tag{2}$$

$$\begin{aligned}\text{3PE signal photons per pulse} &= \frac{S_{3\mathrm{P}}}{f}\\ &= \frac{1}{3}\frac{g_p^{(3)}}{\tau^2}\phi C(\eta\sigma_3)n_0\frac{2\pi^2}{3{\lambda_{3\mathrm{P}}}^3}\\ &\quad\times\left(\frac{\lambda_{3\mathrm{P}}}{\pi w_0}\right)^2\left(\frac{P_{3\mathrm{P}}}{f}\right)^3,\end{aligned} \tag{3}$$

where $g_p^{(2)}=0.66$ and $g_p^{(3)}=0.51$ are, respectively, the second- and third-order temporal coherence factor of the excitation source (assuming a Gaussian temporal profile of the pulse); τ is the laser pulse width (FWHM); ϕ is the system collection efficiency; C is the concentration of the fluorophore; $\eta\sigma_2$ and $\eta\sigma_3$ are, respectively, the 2P and 3P action cross sections; n_0 is the refractive index of the medium; $\lambda_{2\mathrm{P}}$ and $\lambda_{3\mathrm{P}}$ are, respectively, the 2PE and 3PE wavelength in vacuum; w_0 is the $1/e^2$ radius; $\lambda_{3\mathrm{P}}/\pi w_0$ plays the role of the effective NA of a Gaussian beam, defined as n_0 times the $1/e^2$ divergence half-angle of the focus; $P_{2\mathrm{P}}$ and $P_{3\mathrm{P}}$ are, respectively, the average power at the focus for 2PE and 3PE measured in photons/s; and f is the repetition rate.

A. Considerations on Wavelength Selection

The selection of 3PE wavelength affects neuronal imaging in multiple aspects: the 3PE cross section and sensitivity of the calcium indicator are both functions of wavelengths, and the brain heating is determined by the wavelength-dependent tissue scattering and absorption coefficients [Fig. 2(A)].

The optimal 3PE wavelength for GCaMP6 imaging can be derived by quantifying Ca^{2+} imaging with d'. The $\Delta F/F$ of GCaMP6 increases with 3PE wavelength from 1250 to 1375 nm, while the 3PE cross section (and F_0) decreases in the same range [Fig. 4(A)] [43–45]. At the same excitation peak intensity, 1275 nm generates ~70% more signal than 1320 nm but has less than half of the $\Delta F/F$, and the overall effect is that 1275 nm has a lower d'. On the other hand, 1275 nm allows ~50% more average power than 1325 nm [17], which translates to an additional 1.5 times gain in signal. Even with this extra signal taken into account, the overall d' at 1275 nm is still slightly lower than 1320 nm. Therefore, the optimal GCaMP6 imaging wavelength can be determined to center around 1300–1320 nm [43].

1700 nm 3P Ca^{2+} imaging currently faces several challenges. Although redshifted calcium indicators have seen tremendous improvement in recent years and 1700 nm 3P imaging of jRGECOs has been demonstrated in the deep mouse cortex [46], the red genetically encoded calcium indicators are still several times dimmer than GCaMP [47,48]. They also tend to have lower sensitivity after a long period of expression in neurons (e.g., by protein aggregation in lysosomes) [47]. Another limitation is that water absorption is higher with 1700 nm, which results in a lower maximum allowable average power of ~50 mW [7]. With these factors taken into consideration, 1700 nm Ca^{2+} imaging currently can only record a fraction of neurons as 1300 nm 3PM for imaging as deep as the CA1 region in the hippocampus. For a larger imaging depth, 1700 nm 3PM could become advantageous due to the longer EAL [7].

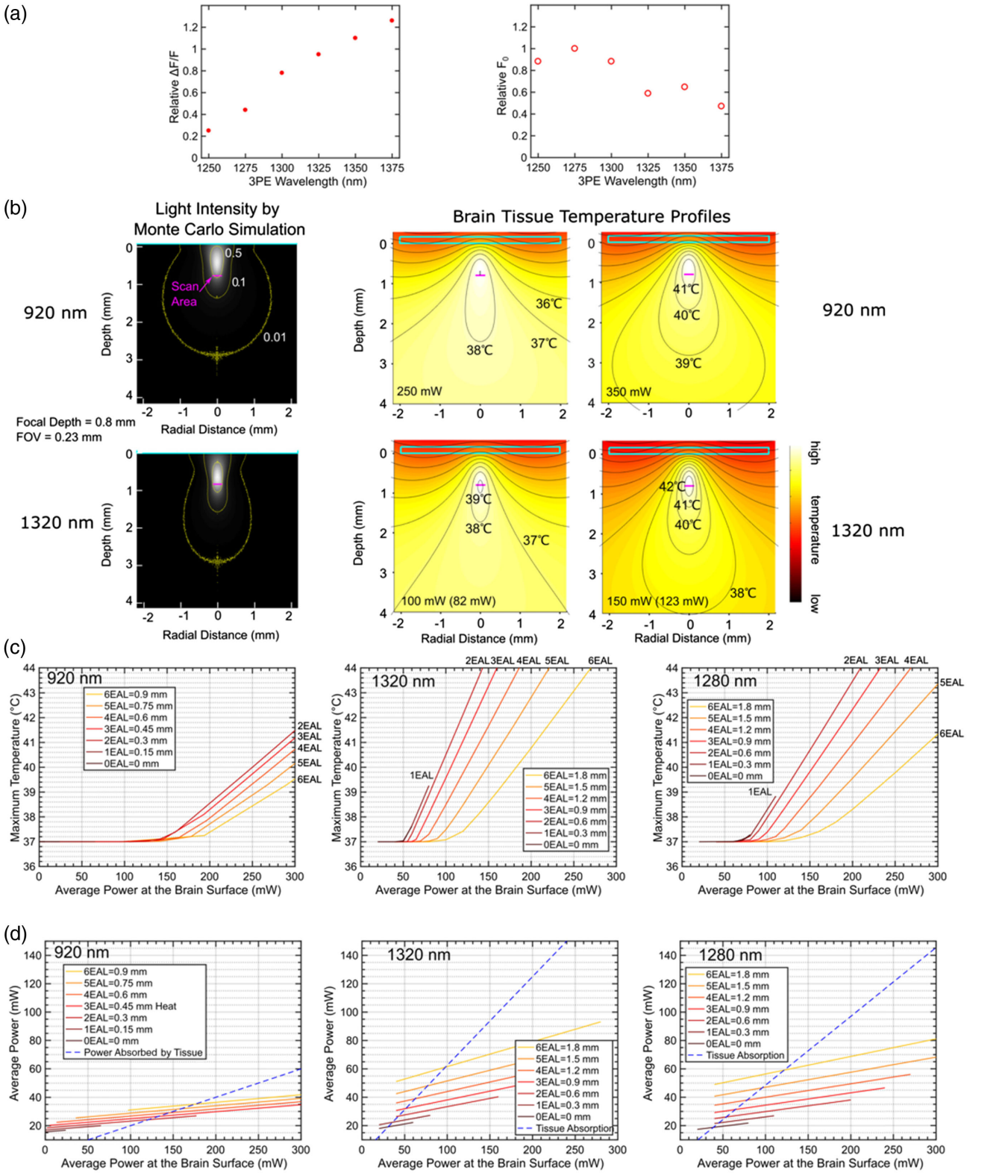


Fig. 4. Effects of 3PE wavelength on GCaMP6 performance and brain tissue heating. (A) The wavelength dependence of the sensitivity and brightness of GCaMP6. Reproduced with permission from Ref. [43], 2019, © The Optical Society. Light intensity distribution and brain temperature profile simulated for 920 nm 2PM and 1320 nm 3PM, after 60 s of continuous scanning (sufficient to reach steady states) at the given average power (the bottom left corner of each plot). For 1320 nm, the distinction is made between the average power after the objective lens and at the brain surface (in the bracket), due to the absorption by the immersion water (H_2O). The absorption by immersion water is negligible at 920 nm, and the power at the brain surface is equal to that after the objective lens. The thickness of the immersion water is calculated as the working distance of the objective (assumed to be 2 mm in these plots) minus the thickness of the cover glass and the imaging depth. (C) The maximum brain temperature as a function of imaging depth and average power, calculated by Monte Carlo simulation for 920, 1320, and 1280 nm. Reproduced with permission from Ref. [17], 2020, eLife Sciences Publications. (D) The power absorbed by the brain tissue (dashed line) and the power dissipated through a 4 mm cranial window (solid lines), simulated for different imaging depth and average power at 920, 1320, and 1280 nm.

B. Power Constraints Imposed by Tissue Heating and Nonlinear Effects

The maximum excitation power for 3PM is limited by laser-induced linear (thermal) and nonlinear effects. Continuous heating leads to increased tissue temperature that perturbs normal physiological functions, including the neuronal firing rate, immune response, and even behavior change [20,49,50]. Compared to 2PM, 3PM increases tissue heating due to increased water absorption at the long wavelengths [51–53]. In addition, the high peak intensity of excitation pulses can adversely affect imaging through fluorophore saturation and nonlinear tissue damage [18,54,55].

Laser-induced brain heating has been systematically studied for 2PM and 1300 nm 3PM [17,20] using both numerical simulation and experimental measurements. A numerical method for simulating brain heating has previously been established and verified by direct measurements in the mouse brain for 2P imaging [20]: the intensity distribution of the excitation light scattered by brain tissue can be predicted by Monte Carlo method, based on which the brain temperature profile can be calculated by solving the thermal diffusion equation [Fig. 4(B)]. The same numerical simulation and immunohistochemistry assay were performed to assess tissue heating in 3P imaging in order to quantitatively relate tissue heating in 3P imaging to that in 2P imaging [17]. For neural imaging in the mouse brain, the cranial window plays a major role in tissue cooling, especially with the overlaying immersion water maintained at room temperature. Without laser illumination, the brain surface temperature can be as low as 32.4°C with cranial window implantation [20,56]. According to the simulation results, the peak tissue temperature only starts to increase above the physiological temperature of 37°C when the heat generated by the excitation light exceeds the cooling capacity of the cranial window [Figs. 4(C) and 4(D)] Above this critical imaging power, the peak tissue temperature increases almost linearly with the average input power [Fig. 4(C)].

The long-wavelength light used by 3PM produces more heat in the brain for the same average imaging power due to the higher tissue absorption [7,17]. Compared to 2PE wavelengths, the 3PE wavelengths have higher tissue absorption coefficients and lower scattering coefficients. As a result, the absorption of the excitation photons become more concentrated within the illuminated/scanned volume, and the excitation photons are more likely to be absorbed than scattered back to the cranial window or escape out of the illumination volume [Fig. 4(B)]. According to Monte Carlo simulation, ∼63% of photons at 1320 nm are absorbed by tissue and contribute to the peak temperature increase in the illuminated volume, in contrast to ∼20% at 920 nm [17]. The higher fraction of photon absorption for 3PM causes the temperature to rise at a lower average imaging power and at a faster rate than 2PM [Fig. 4(C)]. When imaging at 1–1.2 mm imaging depth with 230 μm scanning field of view (FOV), 1320 nm 3PM allows ∼100 mW average power after the objective lens without any thermal effect detectable using immunohistochemistry assays [Fig. 4(A)] [17]. In comparison, the simulation indicates ∼250 mW average power can be used for 920 nm 2PM when scaled to the similar imaging depth and FOV [Figs. 4(B) and 4(C)]. This power limit at 920 nm is similar to that determined by previous experiments at a shallower depth (∼250 μm) but a larger FOV of 1 mm [20]. Figure 4(C) illustrates the simulated tissue peak temperature for imaging depth at various depths. When imaging in other brain regions [13], where the EAL substantially deviates from the typical EALs for the somatosensory cortex used in this simulation (i.e., 150 μm at 920 nm and 300 μm at 1320 nm), the region-specific EAL needs to be used for more accurate temperature prediction since tissue heating depends on the absolute values of the imaging depth and the EAL, not just the ratio of the two (i.e., the EAL-normalized imaging depth).

The higher average power (∼2.5×) allowed by short-wavelength 2PM improves 2P signal strength relative to long-wavelength 3PM; however, this increase in 2P signal does not drastically alter the depth where 3PM is more advantageous than 2PM. At the signal crossover depth of 690 μm (∼2.3 3P EALs; see Section 2.A), the pulse repetition rate of 2PM can be ∼2.5× that of 3PM, which leads to ∼2.5× higher signal strength for 2PM [Eq. (2)]. Assuming the sample is sparsely labeled (e.g., at the volume labeling density $< 0.1\%$) such that 2PM can achieve SBR=1 at greater than 6 times the 2P EAL [22], the higher permissible average power for 2P would push the signal crossover depth from ∼2.3 to ∼3.2 3P EALs. However, in non-sparsely labeled samples, the increase in the 2PE signal is offset by the 2P background. The SBR of 2PM approximately follows the exponential decay of the fluorescence signal. When imaging mouse brain vasculature (volume labeling density ∼2%) or GCaMP6-labeled neurons in transgenic mouse brains (e.g., CamKII-tTA/tetO-GCaMP6s), the SBR decreases ∼4× for every 100 μm increment in depth [17]. According to Eq. (1), a 2.5× increase in F_0 compensates a 1.5× decrease in SBR at the depth where the 2P SBR is ∼1, which only leads to approximately 30 μm additional penetration depth.

Noticeably, the simulation suggests 1280 nm 3PE would allow almost 50% more average power than 1320 nm with a similar EAL, because of the lower water absorption at 1280 nm [51]. Therefore, it may enhance the imaging volume of many green fluorescent molecules, but 1300 nm is still better for GCaMP Ca^{2+} imaging as discussed before. For 1700 nm 3PM, the maximum average power to avoid detectable thermal effects is ∼50 mW, due to the higher water absorption coefficient at 1700 nm [51].

The Monte Carlo simulation discussed above is valid for continuous-wave (CW) illumination. It does not take into account the laser repetition rate or transient tissue heating by a single excitation pulse. Compared to 2PM, 3PM has a lower repetition rate and higher pulse energy at the focus. While the lower laser repetition rate allows heat within the focal volume to largely dissipate (within ∼1 μs) between the pulses [54] and prevents multiple adjacent pulses from having overlapped excitation volumes during scanning, the higher pulse energy increases the transient tissue heating by a single excitation pulse. Based on the water absorption coefficient at 1320 nm ($\mu_a \sim 0.14\ \mathrm{mm}^{-1}$) and the specific heat of water (4.186 J/g°C), a simple calculation using energy conservation and neglecting thermal diffusion shows that a single 1 nJ pulse causes less than 0.1 K temperature rise at the focus for an excitation NA of 1.0 or lower. This value is consistent with a previous calculation for 800 nm laser pulses [54]. Therefore, the transient temperature rise due to a single pulse is small when compared to the steady-state temperature rise at the average power typically used for 3P deep imaging, and Monte Carlo simulation under the CW approximation is valid for predicting the temperature rise of the brain tissue in typical 3PM.

The excitation peak intensity is limited by fluorophore saturation and nonlinear tissue damage. Fluorophore saturation happens

when a large fraction of molecules in the focal volume is excited by a single energetic pulse, which results in lower excitation efficiency and enlarged focal spots [18]. For 1300 nm 3PE of GCaMP6s, approximately 10% of the fluorophores are excited at 2 nJ pulse energy (with 60 fs pulse duration and 0.7 NA focusing). Strong electric field damages healthy tissue through rapid ionization and recombination of molecules [57]. The dependence of this nonlinear damage on the excitation intensity exhibits a threshold-like behavior [55,58]. For neurons labeled with calcium indicators, the damage can be observed as a sudden but irreversible elevation to extraordinary cell brightness [Fig. 5(B)]. In general, nonlinear neuron damage happens at higher pulse energy than 3PE saturation. Empirical data show that neurons remain healthy and viable for weeks after hours of exposure to 1.5 nJ, 60 fs pulses at 1320 nm wavelength [9]. A more recent work assessed neuronal physiology by changing pulse energy at the focal plane for different layers of the cortex, including the white matter [11]. The results showed that, with 40 fs pulse duration and 1.0 NA focusing, at ∼2 nJ or lower pulse energy at the focus, neurons in all layers (including the subplate) of the mouse primary visual cortex exhibit unperturbed orientation tuning to visual stimuli. However, tissue ablation can be occasionally detected when the pulse energy is increased to ∼4 nJ at the focus [Fig. 5(B)]. The threshold for nonlinear effects can be generalized for systems of different parameters by converting the pulse energy to the peak intensity at the focus: $I_{\text{peak}} \propto (P_{3P}/f) \times \text{NA}^2/(\lambda_{3P}^2 \tau)$.

Due to the different mechanisms, heating and nonlinear effects depend on different sets of imaging parameters. Heat-induced damage affects the entire illuminated volume of tissue and depends on the average power, imaging FOV, and depth. On the other hand, nonlinear saturation and nonlinear damage are caused by the peak intensity at the focal point, which is related to focal spot size,

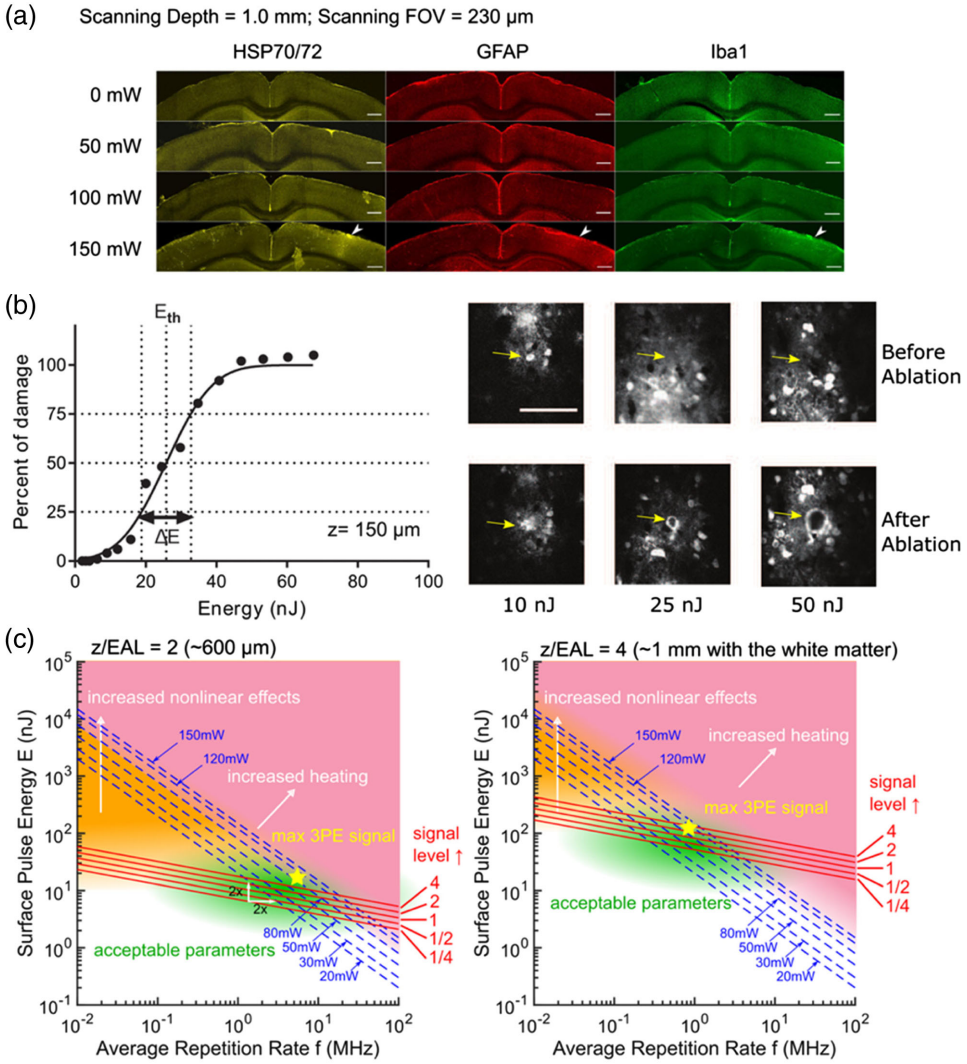


Fig. 5. Optimization of 3P imaging average power and pulse energy within the limits imposed by brain heating and nonlinear effects. (A) Immunostaining results reveal brain heating effects by 1320 nm 3PM after 20 min continuous scanning with 150 mW average power at 1 mm imaging depth. The power indicated in the figure is the average power after the objective lens (with a 2 mm working distance). (B) Tissue ablation induced by intense excitation pulses at the focus. The plot on the left shows the probability of damage (damaged area/total scanned area) versus pulse energy (40 fs pulse duration, 1.0 NA, focused at 150 μm depth). The plot on the right shows the tissue appearance before and after the ablation. Reproduced with permission from Ref. [11], 2019, Springer Nature. (C) Visualization of 3P imaging optimization in the parameter space formed by the pulse energy at the brain surface and the average repetition rate at two imaging depths in the mouse brain. The green region indicates the combinations of the average power and the repetition rate that are allowed and practical for 3P imaging. The yellow star indicates the optimal parameters that maximize the total 3PE signal at the given imaging depth.

pulse energy, and pulse duration. In practice, the high excitation peak intensity at the focus is the primary concern at shallow imaging depth since the average power is low at the repetition rate of several megahertz (MHz; typical of the laser sources for 3PM). As the imaging depth increases, thermal effects caused by the average power in the brain, which grows exponentially with depth, become dominant.

C. Optimization of Three-Photon Neuronal Imaging

The discussion in this section focuses on the 1300 nm spectral window, since the combination of the 1300 nm 3PM and the GCaMPs is currently the most robust and widely adopted approach for 3P Ca^{2+} imaging [9,11,14,59].

The optimization of neuronal imaging entails maximizing d' within the constraints imposed jointly by the sample and instrumentation. Since 3PM is essentially background-free at any practical imaging depth, maximizing the d' of 3PM is equivalent to maximizing the signal [see Eq. (1)]. Because of the cubic power dependence of the 3PE signal, the peak excitation intensity at the focus should be kept as high as possible but below the level at which adverse nonlinear effects (e.g., fluorophore saturation and nonlinear tissue damage) take place. After maximizing the peak intensity, the signal can then be scaled up linearly with the repetition rate until the average power approaches the limit imposed by tissue heating. Figure 5(C) visualizes the optimization procedure in the parameter space. For example, doubling the pulse energy is more effective at enhancing the signal than doubling the repetition rate 8× versus 2× according to Eq. (3), as denoted by the white scale bars in Fig. 5(C), even though the average power is doubled in both cases. The pink and orange regions indicate the combination of parameters that can cause detectable brain heating response and adverse nonlinear effects, respectively. For clarity, only the pink color is displayed in the upper right corner of the graph, even though both heating and nonlinear effects are detectable within this region. The boundary of these regions is not sharp since tissue response is probabilistic in nature, and the power limit imposed by brain heating depends on the scanning FOV. The saturation pulse energy for 3PE in Fig. 5(C) was derived using a 60 fs pulse duration and 0.7 NA focusing.

Figure 5(C) provides an intuitive way to find the optimal repetition rate (f_{opt}) that maximizes the 3PE signal S_{3P} at a given imaging depth z, subject to the constraints on average power and pulse energy. The strongest 3PE signal is achieved when both pulse energy and average power are maximized (i.e., the yellow star at the intersection of the pink, orange, and green regions). Therefore, the optimal repetition rate f_{opt} equals the maximum average power divided by the pulse energy at the brain surface. As the imaging depth increases, the average power outgrows the tissue power-dissipation capacity (exponential versus approximately linear), and therefore the optimal repetition rate decreases with imaging depth [from ∼7 MHz at 600 μm to ∼1 MHz at 1 mm; Fig. 5(C)] [17].

The maximum signal photons per second (S_{3P}^{max}) can be obtained by multiplying the optimal repetition rate f_{opt} with the number of signal photons detected per pulse (S_{3P}/f), assuming a scanning or excitation scheme where each laser pulse lands on a neuron [i.e., only imaging the region of interest (ROI)]. S_{3P}/f can be calculated with Eq. (3). Alternatively, it can also be estimated by scaling with known experimental data. As a reference point, ∼2 nJ at the focus generates 0.1 photons detected per pulse by 1320 nm 3PE of GCaMP6s-labeled neurons, with 0.7 NA focusing and 60 fs pulse duration [17].

The optimal repetition rate fundamentally limits the sampling rate of 3PM and ultimately the number of neurons that can be recorded simultaneously. To estimate the maximum number of neurons that can be recorded, we first calculate the minimum photon counts per second $F_0(d')$ at a given measurement fidelity d' by solving for F_0 in Eq. (1). The maximum number of neurons recorded (N_{max}) is then calculated as $S_{3P}^{max}/F_0(d')$. For example, for GCaMP6s, ∼100 photons per second are needed at the detection accuracy of $d' = 3$ [17]. At 600 μm cortical depth (∼2 EAL), the optimum repetition rate is ∼7 MHz [Fig. 4(C)]. Therefore, N_{max} for 1300 nm 3PM at 600 μm cortical depth and $d' = 3$ is approximately (0.1 photons/pulse) × (7 MHz)/(100 photons/s/neuron) = 7000 neurons. This calculation shows that 3P mesoscopic imaging deep within the mouse brain is feasible if the excitation pulses can be delivered to the ROIs only, complementing the 2P mesoscopic imaging at shallower depth [60–62]. On the other hand, N_{max} decreases approximately by a factor of e for every additional EAL in imaging depth. For imaging the CA1 region of the mouse hippocampus at ∼1 mm depth (approximately 4 to 4.5 EALs due to the presence of the highly scattering external capsule), N_{max} is estimated to be ∼600 to 1000. Figure 6 summarizes the optimization procedure discussed above with a flowchart.

Below we discuss the optimization of excitation pulse duration and focusing schemes in more detail, which are still active areas of research.

3PE benefits more from a shorter pulse width than 2PE. 3PE is inversely proportional to the square of the pulse width, while 2PE is inversely proportional to the pulse width [Eqs. (2) and (3)]. However, extremely short pulses with broad spectrum (e.g., much less than 30 fs with larger than 100 nm spectral bandwidth) may not be advisable due to both practical and fundamental constraints, such as the complication of higher-order dispersion management [63] and the reduction of the excitation efficiency and $\Delta F/F$ due to the inclusion of the less optimal wavelength components [e.g., Fig. 4(A)]. Most 3P imaging has been performed with 35 to 60 fs pulse duration [9,11–14], and more investigation is needed to determine if a much shorter pulse is desirable.

3P Ca^{2+} imaging has been performed with various focal geometry and spot sizes, ranging from 0.5 × 2 to 1.5 × 9.4 μm for the

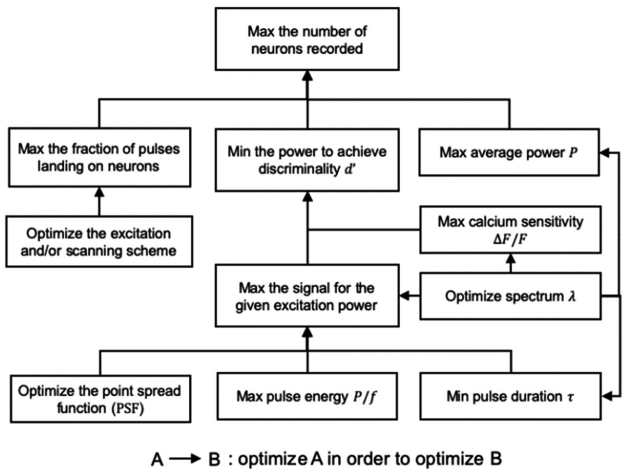


Fig. 6. Three-photon neuronal activity imaging as a constrained optimization problem and the procedure to reach the solution.

lateral and axial FWHM for the 3P PSF [9,11,12,14]. The corresponding focal pulse energy ranges from ∼1 to ∼4 nJ [11,14]. A larger focal spot increases the sampling volume per pulse but reduces the spatial resolution and repetition rate. For deep brain imaging, tissue-induced aberration [39] and increased absorption of the marginal rays within the focusing cone [64] should also be taken into account when imaging with a large NA. In general, the optimal focal geometry and size depend on the applications, and there are opportunities for further quantification and optimization.

4. STRATEGIES FOR VOLUMETRIC THREE-PHOTON NEURONAL IMAGING

The goal for volumetric neuronal imaging is to maximize the number of neurons recorded within limits imposed by tissue heating and nonlinear damage. A number of fast 3D volumetric imaging strategies have been realized for mesoscale 2PM [34,60–62]. Undoubtedly, 3PM substantially benefited from the infrastructures developed for 2PM, including customized optics for large FOV imaging as well as fast lateral and axial scanners; however, not all of the techniques implemented for 2P volumetric imaging are equally translatable to 3PM. Compared to 2PM, 3PM requires higher excitation intensity and permits less average power. In addition, the main advantages of long-wavelength 3PM, when compared to 2PM, are deep imaging (e.g., imaging depths beyond ∼4 EALs, approximately 600 μm in the adult mouse cortex with 920 nm excitation) and imaging through a highly scattering layer (e.g., through the intact mouse skull). Both applications require high power at the brain surface. It is, therefore, critical to maintain the signal generation efficiency while trying to increase the imaging volume rate.

A. Non-Point-Scanning 3P Imaging

3P Wide-Field Imaging with Temporal Focusing: This technique allows 2D wide-field imaging with multiphoton microscopy. The excitation focus is expanded to cover a large area, and the axial resolution is obtained with temporal focusing (TF), which suppresses fluorescence generation away from the focal plane [Fig. 7(A)] [65–67]. Although 700 μm imaging depth has been demonstrated in brain slices with quantum dots [68], wide-field TF is generally not suitable for *in vivo* deep brain imaging. The signal detection of TF requires forming a high-resolution image of the fluorescence emission sources from the focal plane onto a sensor array, such as a 1D or 2D camera. In the presence of strong scattering by the brain tissue at the short fluorescence emission wavelength (e.g., typically less than 700 nm), the lateral resolution and image contrast are severely degraded after multiple scatterings at large imaging depth.

Three-Photon Bessel Beam: The Bessel beam extends the focal depth while maintaining the lateral resolution, and it is capable of acquiring signals from different depths without axial scanning [Fig. 7(A)]. Since the excitation in the axial direction is simultaneous, this technique obtains the axial projection images of the sample. Therefore, it works best in sparsely labeled samples with ideally non-overlapping features at different depths. Mesoscale volumetric 2P imaging has been demonstrated by scanning multiple (up to six) Bessel beams, achieving a 1 Hz volume rate in 3020 μm × 1000 μm × 600 μm volume in the shallow mouse cortex, with a lateral resolution high enough for resolving axonal boutons and dendritic spines [69]. For 3PE, a 60 μm focal depth has been demonstrated *in vivo* in the mouse brain with a Bessel beam, achieving a 3PE volume rate of 1 Hz (50 μm × 50 μm × 60 μm) [Fig. 7(B)] [38]. Compared to 2P Bessel beam imaging, the higher-order excitation of 3PE suppresses the fluorescence signal generated from the side lobes of the Bessel beam, which preserves the lateral resolution and image contrast [Fig. 3(B)] [21]. On the other hand, the inefficient excitation in the side lobes of the Bessel beam, which contains the majority of the excitation power, inevitably leads to low excitation efficiency of the Bessel beam. With 1300 nm excitation at 1 MHz repetition rate, 30–40 mW is required for imaging with a FOV of ∼60 μm located ∼100 μm below the dura [38]. Therefore, this technology is more suited for fast scanning of sparsely labeled fine structure (e.g., dendritic spines) instead of covering a large volume in the deep brain.

Three-Photon Light Sheet Microscopy: Similar to 3PE with a Bessel beam, 3P light sheet microscopy uses either an elongated Gaussian beam or a Bessel beam to scan the sample in the transverse direction. It has been demonstrated that 3PM with 1000 nm excitation penetrates cellular spheroids of ∼450 μm diameter derived from human embryonic kidney cells [70]. However, the high imaging power (∼300 mW) and the transverse geometry of light sheet microscopy makes it challenging for *in vivo* neuronal imaging. In addition, similar to wide-field TF microscopy, imaging of the fluorescence emission using a camera is required, which limits its tissue penetration capability.

The above discussion shows that using the illumination pattern to expand the excitation volume either laterally or axially enabled non-point-scanning 3P imaging to increase the speed of volume image acquisition. The long wavelength and 3PE provide some advantages for improving the image contrast and imaging depth. On the other hand, the increase in 3P imaging volume rate is at the expense of either the excitation efficiency or the imaging resolution and contrast in deep brain imaging. Therefore, compared to point-scanning 3PM, non-point-scanning 3P imaging will be mostly limited to applications within the relatively superficial layers of the brain.

B. Multiplane and Multifoci Imaging

The discussion in this section focuses on axially resolved volumetric imaging techniques enabled by fast axial scanning techniques, in combination with mature lateral scanning methods developed for 2P Ca^{2+} imaging that are equally applicable to 3PM.

Remote Focusing: The remote focusing technique allows rapid movement of the focal plane over tens or even hundreds of micrometers within a few milliseconds [71,72] and has been successfully applied to 2P volumetric imaging [14,60,73]. It has been shown that remote focusing preserves 3PM axial resolution (with ∼0.9 NA focusing) within ∼50−60 μm defocusing distance with off-the-shelf lenses [59]. A dual-plane (each ∼250 μm FOV and at 7 Hz frame rate) and a volumetric imaging scheme (340 μm × 340 μm × 250 μm at 3.9 Hz volume rate) have been implemented to allow simultaneous observation of neuronal populations from different depths [Figs. 7(A) and 7(C)] [14,59]. In particular, the volumetric imaging scheme achieves parallel acquisition from different imaging volumes using time-multiplexed excitation: four small mirrors are employed after the remote focusing objective lens, and each targets a different focal depth. While the setup was implemented for 2PM, it could be extended to 3PM.

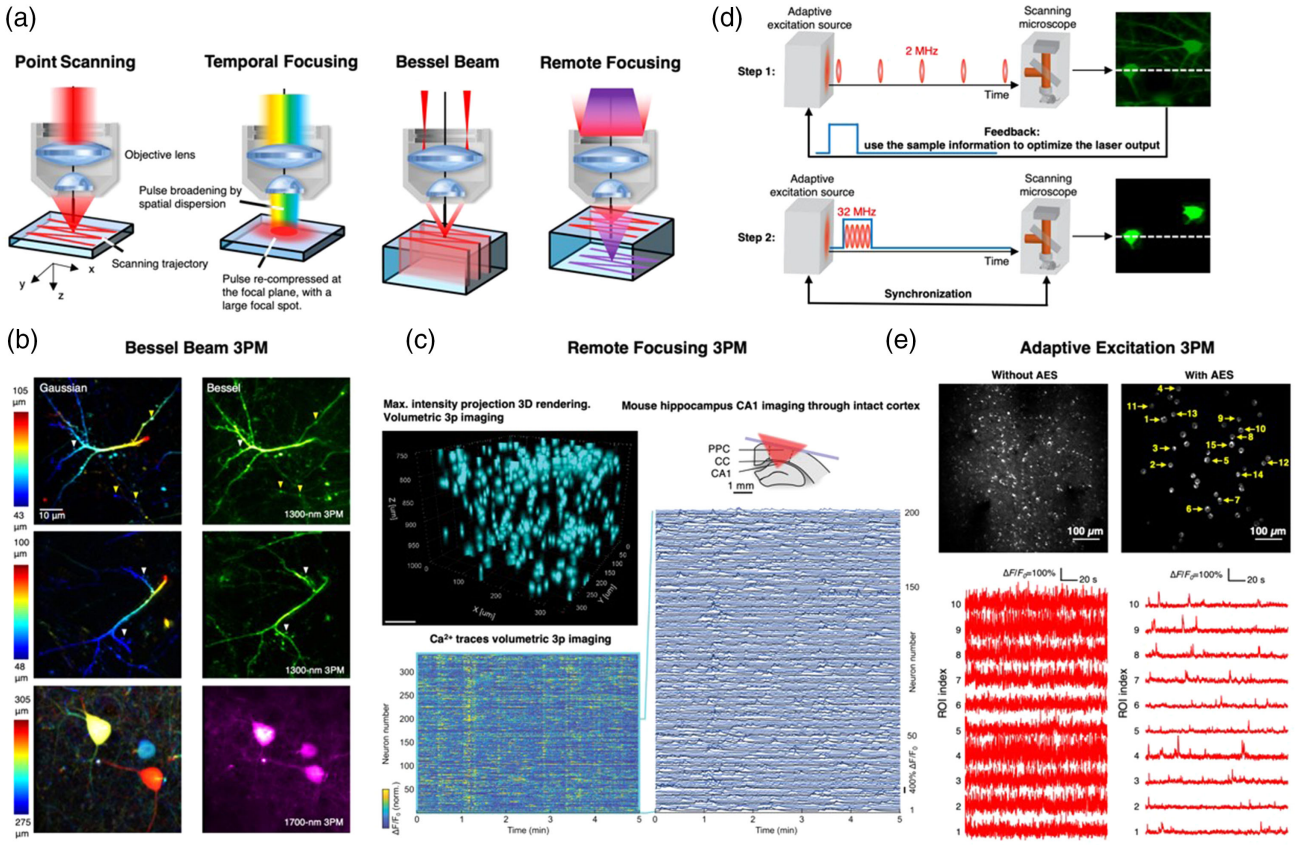


Fig. 7. Strategies for enhancing the imaging volume of 3PM. (A) Illustrations of temporal focusing, Bessel beam, and remote focusing. For temporal focusing, the excitation beam is pseudo-colored to denote the spatial separation of wavelength along the x axis. For remote focusing, two focal depths are illustrated. The focus and the corresponding beam at the objective back aperture are plotted with the same color. (B) *In vivo* volumetric imaging of neurons in the mouse cortex with a Bessel beam. A depth-resolved stack taken by a Gaussian beam is compared to the same stack acquired by a single-frame scan with a Bessel beam. The maximum intensity projection of both stacks is shown, and the imaging depth in the Gaussian beam stack is color-coded. Reproduced with permission from Ref. [38], 2018, © The Optical Society. (C) High-speed and volumetric 3PM imaging of GCaMP6f-labeled neurons in the mouse hippocampus through the intact cortex. The recording depth was 750–1000 mm, with 340 μm × 340 μm × 250 μm imaging volume and 3.9 Hz volume rate. Reproduced with permission from Ref. [14], 2019, Elsevier. (D) The working principle of an adaptive excitation source (AES). (E) 1700 nm 3P Ca^{2+} imaging of jRGECO1a-labeled neurons 750 μm below the brain surface with an AES, taken at 620 μm × 620 μm FOV at 30 Hz frame rate. Reproduced with permission from Ref. [46], 2020, Springer Nature.

Dual-wavelength time-multiplexed remote focusing also opens up a new opportunity of simultaneous imaging with 2PM and 3PM in the shallow and deep brain, respectively [14,74]. By placing a longpass dichroic mirror (1300 nm pass and 920 nm reflect) between the remote focusing objective and the mirror, it is possible to tune the 2P and 3P imaging planes independently with a single remote focusing module [74]. Overall, remote focusing is the most mature technique that offers axial scanning capacity in 3PM.

Reverberation Microscopy: Reverberation microscopy achieves similar functionality as the multimirror remote focusing approach discussed above [14], but with a more scalable and simpler implementation [75]. The "reverberation loop" splits an energetic pulse traveling inside into a train of output pulses equally spaced in time. By adding a lens in the loop, the spatial divergence of each successive pulse within the pulse train varies monotonically, allowing the pulses to focus at different imaging depths. The power of each pulse leaving the loop decays as a geometric series that naturally scales with the power required at the target imaging depth for each pulse [75]. Although not yet reported for 3PM, this device potentially offers an integrated and scalable solution to time-multiplexed multiplane imaging with ∼10 ns plane-to-plane switch time.

C. Adaptive Excitation Sources

Ultrafast lasers are the engines for multiphoton microscopy. Pushing the boundaries of imaging performance often requires the development of new excitation sources that did not exist before. The deep imaging capability of 3PM spurred commercial and research development [76,77] of laser sources in the last five years. Today, femtosecond lasers for 3PM are available from a number of vendors. These lasers are robust and energetic [∼microjoule (μJ) pulse energy], with tunable repetition rate [e.g., from hundreds of kilohertz (kHz) to several megahertz (MHz)] and tunable wavelength (e.g., covering both 1300 nm and 1700 nm spectral windows).

In an average-power-limited situation, which is typical for deep brain imaging, an effective strategy to increase the imaging speed is to concentrate all the excitation pulses on the neurons for recording neural activity [Fig. 7(D)]. An adaptive excitation source (AES) delivers laser pulses on demand to the neuron ROIs,

based on the prior knowledge of neuron locations obtained with a full scanning of the region [46]. The AES can be achieved by temporally modulating a seed pulse train of a high repetition rate. By selectively picking and then amplifying the pulses, on-demand delivery of the excitation pulses onto the ROIs can be achieved by synchronizing the laser pulse train with the scanner [Fig. 7(D)]. Since neurons only occupy a small fraction of the imaged volume, the average repetition rate is kept low to avoid brain heating. It has been shown that AES achieves the same signal strength with ∼1/30 average power for 1700 nm 3PM imaging of jRGECO1a-labeled neurons at 750 μm deep in the mouse cortex [Fig. 7(E)] [46]. In other words, by simply repositioning the pulses in time and synchronizing the pulses with the scanner, the AES allows 30 times larger scanning FOV and the number of neurons recorded by 3PM without increasing the average power and peak power in the brain or from the laser. Similar performance improvement using the AES was also demonstrated for 2PM. As estimated in Section 3.C., the AES can be combined with a mesoscale imaging system [60–62] to record many thousands of neurons in the deep cortex. Sensitivity to motion artifacts is a major drawback for any method that only images the ROIs, including the AES. Enlarging the ROIs is an effective way to overcome sample motion, but it reduces the overall power efficiency. A single AES is incompatible with simultaneous multifoci or multiplane imaging without incurring the penalty on power efficiency. If fast z-scanning is deployed, however, nearly simultaneous (in the context of the biological dynamics) imaging of multiple axial planes can be achieved by the AES.

More generally, the AES represents the concept of adaptive tuning of the laser parameters according to the experimental settings and sample conditions, which is a promising new direction for further optimization of ultrafast laser sources for 2P and 3P imaging. Indeed, much of the optimization requires tuning of the laser repetition rate. Since the maximum average power permitted does not vary dramatically as a function of depth for deep brain imaging, an ideal, efficient excitation source should allow tunable repetition rate while maintaining constant average output power, i.e., as the repetition rate decreases, the pulse energy will increase accordingly. Such a laser can maintain the pulse energy at the focus when imaging deep and the average power at the brain surface (limited by sample heating). The repetition rate of a mode-locked femtosecond laser is defined by the cavity length and typically cannot be tuned over a broad range (e.g., 1 to 10 MHz). Excitation sources based on chirped pulse amplification systems (e.g., optical parametric chirped pulse amplifier, OPCPA), which include most of the 3P sources, can provide a tunable repetition rate. However, because of the optical nonlinearity encountered in the pump amplifier or the subsequent wavelength conversion process, the pulse energy of the commercially available OPCPA systems is typically fixed within a narrow range regardless of the pulse repetition rate. While the repetition rate is tunable, these sources only allow optimum performance over a narrow range of imaging depth (defined by its maximum pulse energy). For example, imaging shallower than the source is designed for, the repetition rate of the source is too low, and the pulse energy is too high. One must attenuate the pulse energy (i.e., wasting precious optical power) and image slower than the optimum. External doubling of the repetition rate can be performed by using a long optical delay line [9,14], but the setup is cumbersome and provides only a limited tuning range. Ultrafast lasers are the engines for multiphoton imaging since its inception 30 years ago. Future development on ultrafast lasers, such as tunable repetition rate lasers or pulse-on-demand systems (e.g., the AES), has the potential to transform nonlinear microscopy.

5. SUMMARY AND PERSPECTIVE

In this review, the advantages and disadvantages of long-wavelength 3PM have been presented by comparing it to 2PM. The optimization of the performance of 1300 nm 3P Ca^{2+} imaging has also been discussed in detail. The same quantitative analysis can be applied to voltage indicators and redshifted calcium indicators with 1700 nm 3PE. The promise and the limitation of 3PM for large-scale, volumetric neuronal imaging have been illustrated based on the fundamental physical constraints.

The development of long-wavelength 3PM complements 2PM for deep tissue imaging. Indeed, the advantages of 3PM derive from the same basic principles that enabled 2PM to become such a powerful tool for imaging intact tissues: long-wavelength excitation for reducing the effects of tissue scattering and nonlinear excitation for producing a tightly confined excitation volume (i.e., a sharp focus). It is pure coincidence that water, which is by far the most dominant component of biological tissues *in vivo*, is relatively transparent at 1300 nm and 1700 nm, and these two spectral windows match well with 3PE of existing blue/green and orange/red fluorophores. Without the low tissue absorption in these spectral windows, long-wavelength 3PM and its applications would be severely limited by tissue heating.

Would four-photon excitation (4PE) enable deeper imaging? Simple estimations based on quantum perturbation theory and measurements have shown that 4PE is feasible, and four-photon microscopy (4PM) has been demonstrated in the past [19]. On the other hand, the advantage of 4PM over 3PM in terms of deep tissue penetration is questionable. As discussed before, the SBR does not limit 3PM at any practical imaging depth. Therefore, better excitation confinement by 4PE does not bring tangible benefits for deep imaging. When compared to 3PE, 4PE would require a longer excitation wavelength; while reducing the effects of tissue scattering even further, tissue absorption increases dramatically beyond 1850 nm [51,53]. Given the higher-order nonlinear excitation (lower excitation efficiency), it would be challenging to find the parameter space (e.g., wavelength, repetition rate, pulse energy, etc.) where 4PM outperforms 3PM in the mouse brain. On the other hand, 4PE does expand the spectral access to fluorophores (e.g., 4PE of green fluorescent protein at 1700 nm), and the reduction of scattering can be important in other tissues with different scattering and absorption properties. Further investigation of 4PM in various biological tissues is needed to determine its value for deep tissue imaging.

High-spatial-resolution fluorescence imaging in deep scattering tissue is challenging because the "difficulty" grows exponentially as a function of imaging depth. Long-wavelength 3PM overcomes some of the limitations for 2PM, but much deeper imaging is certainly desired. Although the imaging depth of fluorescence microscopy has increased substantially in the last three decades, mostly driven by the innovations of multiphoton microscopy, even the deepest 3P imaging cannot penetrate a quarter of an adult mouse brain *in vivo*. While the significance of deep tissue imaging cannot be overstated, it remains an open challenge on how to image much deeper than the current state of the art.

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